

Name: _____

Score: _____ / _____

1 Examination 1

UNIVERSITY EXAMINATIONS

UNIVERSITEITSEKSAMENS



MAT1503

May/June 2020

Linear Algebra I

Duration : 2 Hours

100 Marks

EXAMINERS :

FIRST : DR ZI ALI
SECOND : DR M DAVIES

Closed book and online examination, which you have to write within 2 hours and submit online, by no later than 17:00 on 25 June 2020.

This web based examination remains the property of the University of South Africa and may not be distributed from the Unisa platform.

Declaration: I have neither given nor received aid on this examination.

The examination is made up of multiple-choice question and it will take place on the Online Assessment tool located on your module examination site on myUnisa. You have to be online for the full duration of the exam.

Save Frequently while working

Answer All Questions and Submit within the stipulated timeframe.

Late submission will not be accepted.

Part 1

Question 1 - [4 Marks]

Let C be an m by n matrix and k a scalar such that $kC = 0$. Which of the following statements is/are correct?

A. Either $k = 0$ or $C = 0$.

B. k is not zero and C is not the zero matrix.

☐ A.

Only A.

☐ B.

☐ C.

Both A and B.

☐ D.

Neither A nor B

☐ E.

Only B.

Question 2 - [4 Marks]

Let T be an m by n matrix.

Which of the following statements is/are correct?

A. $T = -T$ if and only if T is the m by n zero matrix.

B. $T = -T$ if T is an $m \times n$ symmetric matrix.

- ☐ A.
Neither A nor B.
- ☐ B.
Only B.
- ☐ C.
Only A.
- ☐ D.
Both A and B.

Question 3 - [6 Marks]

Let A and B be 3×3 matrices. Given that

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 3 & 2 & 0 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{bmatrix}.$$

Which of the following statements is/are correct?

$$\text{C. } (AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} \frac{8}{3} & -1 & -\frac{1}{3} \\ -2 & 1 & 0 \\ \frac{5}{3} & -1 & \frac{2}{3} \end{bmatrix}$$

$$\text{D. } (BA)^{-1} = A^{-1}B^{-1} = \begin{bmatrix} 1 & -2 & -3 \\ 0 & 1 & 0 \\ -\frac{2}{3} & \frac{4}{3} & \frac{7}{3} \end{bmatrix}$$

- ☐ A.
Only C.
- ☐ B.
Neither C nor D.
- ☐ C.
Only D.
- ☐ D.
Both C and D.

Question 4 - [7 Marks]

Use Gaussian elimination to solve the following system of equations.

$$\begin{cases} 2x + 2y - 4z = 0 \\ 2x + 2y - z = 1 \\ 3x + 2y - 3z = 3 \end{cases}$$

Which of the following statements is/are correct?

A. $x = 8, y = -2, z = 1$ is the unique solution.

B. $x = -\frac{8}{3}, y = 2, z = \frac{1}{4}$ is the only solution.

- ☐ A.
- ☐ Both A and B.
- ☐ B.
- ☐ Neither A nor B.
- ☐ C.
- ☐ Only B..
- ☐ D.
- ☐ Only A.

Question 5 - [5 Marks]

Let C and D be two square matrices of the same size. Let E be the inverse of D . If $\det(C) = 3$ and $\det(D) = 4$, what is the value of $\det(CE)$?

A. $\det(CE) = 12$.

B. $\det(CE) = \frac{3}{4}$.

- ☐ A.
Only B.
- ☐ B.
Only A.
- ☐ C.
Neither A nor B
- ☐ D.
Both A and B.

.

Question 6 - [4 Marks]

Consider the following system of equations

$$\begin{cases} x + y = 1 \\ x + z = 2 \\ y + z = 3 \end{cases}$$

What is the matrix form of the system, *i.e.*, $AX = B$?

C. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 9 \end{bmatrix}$

D. $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

- ☐ A.
Neither C nor D.
- ☐ B.
Only D.
- ☐ C.
Only C.
- ☐ D.
Both C and D.

Question 7 - [6 Marks]

Consider the following system of equations

$$\begin{cases} x + y = 1 \\ x + z = 2 \\ y + z = 3 \end{cases}$$

Which of the following statements is/are correct?

C. The system has exactly one solution if

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

D. The system has exactly one solution if A is singular

- ☐ A.
- ☐ Both C and D.
- ☐ B.
- ☐ Only C.
- ☐ C.
- ☐ Only D.
- ☐ D.
- ☐ Neither C nor D.

Question 8 - [5 Marks]

Consider the following system of equations

$$\begin{cases} x + y = 1 \\ x + z = 2 \\ y + z = 3 \end{cases}$$

If A is the coefficient matrix of the above system; then using the inverse of given by

$$A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Deduce which of the following statements is/are correct?

C. The solution of the system is $X = (0, -1, 2)$.

D. The solution of the system is given by $x = 1, y = 0, z = 2$

- ☐ A.
Only D.
- ☐ B.
Only C.
- ☐ C.
Neither C nor D.
- ☐ D.
Both C and D.

Question 9 - [5 Marks]

Given the following system of equations

$$\begin{cases} 2x + 3y - z = 1 \\ 4x + y + 2z = 5 \\ x + y + z = 2 \end{cases}$$

Find A_x , A_y , A_z from which calculate $\det(A_x)$, $\det(A_y)$, and $\det(A_z)$.

Which of the following statements is/are correct?

C. $\det(A_x) = -7$, $\det(A_y) = -3$ and $\det(A_z) = -12$.

Then $x = \frac{\det(A_x)}{\det(A)} = \frac{7}{11}$, $y = \frac{3}{11}$, $z = \frac{12}{11}$.

D. $\det(A_x) = -7$, $\det(A_y) = -3$ and $\det(A_z) = -12$.

Then $x = \frac{\det(A_y)}{\det(A)} = \frac{3}{11}$, $y = \frac{7}{11}$, $z = \frac{12}{11}$.

- ☐ A.
Only D.
- ☐ B.
Only C.
- ☐ C.
Both C and D.
- ☐ D.
Neither C nor D.

Question 10 - [4 Marks]

Let A be an n by n matrix. Suppose that in the system $A X = B$ all of the coefficients and constant terms are integers. Here B is a column vector.

Which of the following statements is/are correct?

C. If the solutions are all integers then $\det(A) = 1$.

D. The converse of the statement in C is true.

☐ A.
Neither C nor D.

☐ B.
Only D.

☐ C.
Only C.

☐ D.
Both C and D.

Question 11 - [6 Marks]

Consider the vector $\underline{u} = (1, 0, 4)$ and $\underline{v} = (1, 1, 0)$ in standard position.

What is the orthogonal complement of $\underline{v}$ onto $\underline{u}$?

A. $X = \left(\frac{2}{17}, 0, \frac{3}{17}\right)$.

B. $Y = \left(\frac{1}{17}, 0, \frac{3}{17}\right)$.

- ☐ A.
Neither A nor B.
- ☐ B.
Only B,
- ☐ C.
Only A.
- ☐ D.
Both A and B.

Question 12 - [4 Marks]

Consider the vector $\underline{u} = (1, 0, 4)$ and $\underline{v} = (1, 1, 0)$ in standard position.

Which of the following statements is/are correct?

A. The area of the parallelogram bounded by $\underline{u}$ and $\underline{v}$ is $\sqrt{33}$ unit².

B. The area of the parallelogram bounded by $\underline{u}$ and $\underline{v}$ is $\sqrt{7}$ unit².

- ☐ A.
Neither A nor B.
- ☐ B.
Only A.
- ☐ C.
Both A and B.
- ☐ D.
Only B.

Question 13 - [4 Marks]

Consider the vector $\underline{u} = (1, 0, 4)$ and $\underline{v} = (1, 1, 0)$ in standard position.

Find the perimeter of the parallelogram bounded by $\underline{u}$ and $\underline{v}$. Which of the following statement(s) is/are correct?

A. The perimeter of the parallelogram is *21 m*.

B. The perimeter of the parallelogram is *19 m*.

- ☐ A.
Only B.
- ☐ B.
Only A.
- ☐ C.
Both A and B.
- ☐ D.
Neither A nor B.

Question 14 - [5 Marks]

Find an equation of the plane containing the point $(0, 1, 1)$ and perpendicular to the line passing through the points $(2, 1, 0)$ and $(1, -1, 0)$.

Which of the following statements is/are incorrect?

A. The equation of the plane is $x + 2y = 2$.

B. The equation of the plane is $-x - 2y = -2$.

- ☐ A.
- ☐ Both A and B.
- ☐ B.
- ☐ Only B.
- ☐ C.
- ☐ Only A.
- ☐ D.
- ☐ Neither A nor B.

Question 15 - [4 Marks]

Find the parametric equations of the line that passes the point $(2, 3, 4)$ and is perpendicular to the plane $x + 2y - z = 0$.

Which of the following statements is/are correct?

A. If the direction vector is $\underline{v} = (-1, -2, 1)$ then the parametric equations of the line are $x - t = 2, y - 2t = 3, z + t = -4$.

B. If the line passes through the point $(2, 3, 4)$ then the parametric equations are

$$x - 2 = t,$$

$$y - 3 = 2t,$$

$$z - 4 = -t.$$

- ☐ A.
Only A.
- ☐ B.
Only B.
- ☐ C.
Both A and B.
- ☐ D.
Neither A nor B.

Question 16 - [3 Marks]

Solve for z given that $z + (1 - i) = 3 + 2i$.

Which of the following statements is/are correct?

A. $z = 2 + 3i$.

B. $z = 1 - i$ or $z = 3 + 2i$.

- ☐ A.
Only B.
- ☐ B.
Neither A nor B.
- ☐ C.
Only A.
- ☐ D.
Both A and B.

Question 17 - [4 Marks]

Express

$$\frac{i}{i+1} \text{ in the form } a + ib, a, b \in \mathbb{R}$$

Which of the following answer(s) is/are the simplified correct answer(s) for the above form?

A. $i - 1$.

B. $\frac{1}{2} + \frac{1}{2}i$.

☐ A. B. $\frac{1}{2} + \frac{1}{2}i$.

☐ B. Both A and B.

☐ C. Neither A nor B.

☐ D. A. $i - 1$.

Question 18 - [8 Marks]

Use De Moivre's Theorem to express

$(\sqrt{3} + i)^8$ in the form $a + ib$, $a, b \in \mathbb{R}$

Which of the following statements is/are incorrect?

A. $\arg(\sqrt{3} + i)^8 = \frac{\pi}{6}$ and $(\sqrt{3} + i)^8 = 2^7 (1 + i)$.

B. $(\sqrt{3} + i)^8 = 2^8 \left(\frac{1}{2}i + \frac{\sqrt{3}}{2} \right)$.

- ☐ A.
- ☐ Both A and B.
- ☐ B.
- ☐ Only B.
- ☐ C.
- ☐ Neither A nor B.
- ☐ D.
- ☐ Only A.

Question 19 - [4 Marks]

Find the cube roots of -1 in the form $a + i b$.

Which of the following answer(s) is/are the simplified correct answer(s) for the above form?

A.

$$z_0 = \cos \pi - i \sin \pi = -1 - 0i,$$

☐ $z_1 = \cos(2\pi) + i \sin(2\pi) = 1 + 0i,$

$$z_2 = 2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right) = 1 + \sqrt{3}i.$$

B.

$$z_0 = 2(\cos \pi - i \sin \pi) = 2 + 0i,$$

☐ $z_1 = \cos \pi + 2i \sin \pi = -1 + 0i,$

$$z_2 = 3 \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right) = 0 - 3i.$$

C.

$$z_0 = \cos \pi - i \sin \pi = -1 - 0i,$$

☐ $z_1 = \sqrt{2} \left[\cos \left(\frac{\pi}{4} \right) + i \sin \left(\frac{\pi}{4} \right) \right] = 1 + i,$

$$z_2 = \sqrt{2} \left[\cos \left(\frac{3\pi}{4} \right) + i \sin \left(\frac{3\pi}{4} \right) \right] = -1 + i.$$

D.

$$z_0 = \cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i,$$

○ $z_1 = \cos \pi + i \sin \pi = -1 + 0i,$

$$z_2 = \cos \left(\frac{5\pi}{3} \right) + i \sin \left(\frac{5\pi}{3} \right) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

Question 20 - [4 Marks]

Eliminate θ from the equations

$$4x = \cos(3\theta) + 3\cos\theta;$$

$$4y = 3\sin\theta - \sin(3\theta)$$

Which of the following statement(s) is /are incorrect?

A. $x^2 + y^2 = 1.$

B. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 2$

- ☐ A.
Only B.
- ☐ B.
Only A.
- ☐ C.
Both A and B.
- ☐ D.
Neither A nor B.

Question 21 - [4 Marks]

Express $\cos(4\theta)$ and $\sin(5\theta)$ as polynomials in terms of powers of $\cos \theta$ and $\sin \theta$.

Which of the following statement(s) is/are correct?

- A. $\cos(4\theta) = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta,$
 $\sin(5\theta) = \sin^5 \theta - 10 \sin^3 \theta \cos^2 \theta + 5 \sin \theta \cos^3 \theta.$
- B. $\cos(4\theta) = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta,$
 $\sin(5\theta) = \sin^5 \theta - 15 \sin^3 \theta \cos^2 \theta + 5 \sin \theta \cos^3 \theta.$

- A.
- ☐ Only A.
- B.
- ☐ Only A.
- C.
- ☐ Both A and B.
- D.
- ☐ Neither A nor B.

Question 22 - [3 Marks]

Let

$$4x = \cos(3\theta) + 3\cos\theta.$$

Express $\cos(4\theta)\sin(3\theta)$ in terms of multiple angles and in term of x .

Which of the following statement(s) is/are correct?

A. $\cos(4\theta)\sin(3\theta) = 2^7 x [\cos(4\theta) - 4\cos(2\theta) + 3].$

B. $\cos(4\theta)\sin(3\theta) = 2^7 x^2 [\cos(4\theta) - 4\cos(2\theta) + 3].$

- A.
- ☐ Only A.
- B.
- ☐ Neither A nor B.
- C.
- ☐ Both A and B.
- D.
- ☐ Only B.